

APL – Amharic Programming Language Project Report

The purpose of this project was to design a small programming language that makes programming more approachable for beginners by using an Ethiopian local language. I selected Amharic because it is widely spoken and familiar to many students who may encounter programming for the first time.

The main design decision was replacing common English programming keywords with Amharic equivalents. For example, if became ከሆነ, while became ሲሆን, and return became መልስ. This allows learners to focus on programming logic without simultaneously struggling with unfamiliar English syntax.

Another important design choice was the introduction of explicit block markers (ጀምር and መጨረሻ). Many beginner errors in languages such as Python are caused by incorrect indentation. Using visible block delimiters makes program structure easier to understand and simplifies the parser implementation.

The translator was implemented using three separate stages: lexical analysis, parsing, and code generation. The lexer converts source code into tokens, the parser builds an Abstract Syntax Tree (AST), and the code generator produces valid Python code. This ensures that the project performs genuine source-to-source translation rather than simple keyword replacement.

One of the biggest challenges was supporting Unicode characters correctly. Since both keywords and identifiers may contain Amharic characters, the lexer had to recognize Unicode text reliably. Designing a grammar that remained simple while supporting functions, loops, conditionals, and variable assignments was another challenge.

Overall, the project demonstrates that local-language programming is both feasible and educationally useful. APL successfully supports variables, arithmetic expressions, comparisons, conditional statements, loops, functions, and return values while maintaining a syntax that is easier for Amharic-speaking beginners to read and understand. This approach may help reduce the initial barriers to learning programming and encourage greater participation among new learners.